

1. State the definition of a group.
2. Write the addition table for the group $\mathbb{Z}_3$.
3. Give an example of a group with infinitely many elements, having the property that every nontrivial subgroup also has infinitely many elements.
4. State three different abelian groups with 8 elements. (You do not have to write out their tables.)
5. **Optional extra credit:** State two different *nonabelian* groups with 8 elements.